1.Open example.com in your browser, press F12, and use the Elements panel to change the h1 text to the name of a tuition class you invent, rewrite the first paragraph as a sentence of at least 10 words, and change the page background colour. Save a screenshot named 'dom_edit.png' showing the edited page with DevTools open. Hint: Edits last only until you reload, so take the screenshot first; double-click text in the Elements panel to edit it. Constraint: Make all edits inside DevTools; do not download or copy the page.

Step-by-step instructions

1. Open the website

1. Open Chrome or Edge.
2. Visit https://example.com
3. Press F12 to open Developer Tools.
4. Select the Elements tab.

2. Change the heading (h1)

1. Find the <h1> element.
2. Double-click its text.
3. Replace it with:

Bright Future Tuition Classes

3. Change the first paragraph

Find the first <p> element and replace its text with:

We provide quality education and personal guidance to help every student achieve academic success.

(This is more than 10 words.)

4. Change the background colour

1. Select the <body> element in Elements.
2. In the Styles panel, find or add element.style.
3. Add:

background-color: lightblue;

5. Save your screenshot

GitHub - pankajparashar/canidev.tools: It is like @CanIUse but for the browser devtools. Created by @pankajparashar & curated by community. · GitHub

github.com

1. Keep the edited webpage visible.
2. Keep DevTools open.
3. Press Windows + Shift + S.
4. Capture the browser and DevTools together.
5. Save the screenshot as:

dom_edit.png

Final checklist

- Heading changed to Bright Future Tuition Classes.
- First paragraph rewritten with at least 10 words.
- Background changed to light blue.
- DevTools Elements panel visible.
- Screenshot saved as dom_edit.png.

2. Create a text file named 'dom_tree.txt' that writes the DOM tree of example.com as indented tag names, starting at html and including head, body and every tag inside them, with two spaces for each level of nesting. Hint: In the Elements panel, click the small triangle beside an element to expand it and see its children. Constraint: Write each tag once for every time it appears on the page.

1. DOM tree of example.com

Open example.com → press F12 → select Elements.

The default example.com page has this DOM structure:

File name: dom_tree.txt

```
html
 head
  title
 body
  div
   h1
  p
  p
   a
```

2. Explanation

- html – Root element of the webpage.
- head – Contains page information.
- title – Contains the webpage title.
- body – Contains visible webpage content.
- div – Main content container.
- h1 – Main heading.
- p – Paragraph elements.
- a – Link inside the second paragraph.

3. Save the file

1. Open Notepad.
2. Copy the DOM tree above.
3. Click File → Save As.
4. Enter dom_tree.txt.
5. Select Save as type: All Files.
6. Save your file.

Final checklist

- File named dom_tree.txt.
- Starts with html.
- Includes head and body.
- Two spaces for each nesting level.
- Every tag appears as many times as it occurs in the DOM.

3. Create a Draw.io file named 'browser_flow.drawio' that draws, as at least five boxes joined by numbered arrows, how a browser turns the address you type into a painted page, including the HTML and CSS being received and the DOM being built. Hint: When Draw.io asks where to save, choose Device. Constraint: Label every arrow with what travels along it, such as a request or an HTML file.

1. Open Draw.io

1. Visit Draw.io.
2. Select Device when asked where to save.
3. Choose Create New Diagram.
4. Name it browser_flow.
5. Select Blank Diagram → Create.

2. Create the browser flowchart

Add these six boxes and connect them using numbered arrows.

1. User enters URL

example.com

① HTTP request / DNS lookup

2. DNS and Web Server

Finds the server IP address

② HTTP request

3. Server sends files

HTML file and CSS file

③ HTML and CSS responses

4. Browser parses HTML

Builds the DOM tree

④ CSS rules and DOM

5. CSS processing and layout

Builds CSSOM and calculates positions

⑤ Render information

6. Browser paints the page

Displays the final webpage

3. Save your file

1. Click File → Save.
2. Choose Device if asked.
3. Save as browser_flow.drawio.

Task 3 checklist

- At least five boxes.
- Six numbered steps.
- Every arrow has a label.
- HTML and CSS receiving included.
- DOM building included.
- Final browser painting included.
- File saved as browser_flow.drawio.

4. Finish by adding a second page named 'Front and Back' to 'browser_flow.drawio' that sorts ten parts of a tuition-class website into two boxes, Front-end and Back-end, then push 'dom_edit.png', 'dom_tree.txt' and 'browser_flow.drawio' to a new public GitHub repository named 'web-basics-lab'. Constraint: Put exactly five parts in each box, each written as a specific feature such as 'enquiry form' or 'student records'.

Part 1: Add the second page in Draw.io

Open your existing browser_flow.drawio file.

1. Open Draw.io.
2. Choose Device and open browser_flow.drawio.
3. At the bottom, click the + icon to add a new page.
4. Rename the new page to Front and Back.
5. Add two large rectangles side by side.

Use exactly these ten features:

Front-end

5 features

1. Tuition class homepage
2. Student registration form
3. Enquiry form
4. Course details page
5. Contact information section

Back-end

5 features

1. Student records database
2. Student login authentication
3. Fee payment processing
4. Attendance management
5. Exam results management

6. Save using Ctrl + S.

Your Draw.io file now contains two pages:

- Page 1: Browser Flow
- Page 2: Front and Back

Part 2: Push files to GitHub

GitHub Vs. Bitbucket: Which Code Hosting Platform is Best? - Hatica

Create a public repository

Open GitHub – Create a new repository.

Use these settings:

Setting	Value
Repository name	web-basics-lab
Description	Web Basics Lab Assignment
Visibility	Public
Add README	Optional

Click Create repository.

Upload your three files

On the new repository page:

1. Click Add file.
2. Select Upload files.

3. Select these three files from your computer:

dom_edit.png

dom_tree.txt

browser_flow.drawio

4. Scroll down.
5. Enter commit message: Add web basics lab assignments.
6. Click Commit changes.

Final submission checklist

- browser_flow.drawio has two pages.
- Second page is named Front and Back.
- Front-end has exactly five features.
- Back-end has exactly five features.
- GitHub repository is named web-basics-lab.
- Repository visibility is Public.
- All three files uploaded.

GitHub connection: I've suggested the GitHub integration, which can help publish your work. You need to connect it before I can help with repository operations directly.